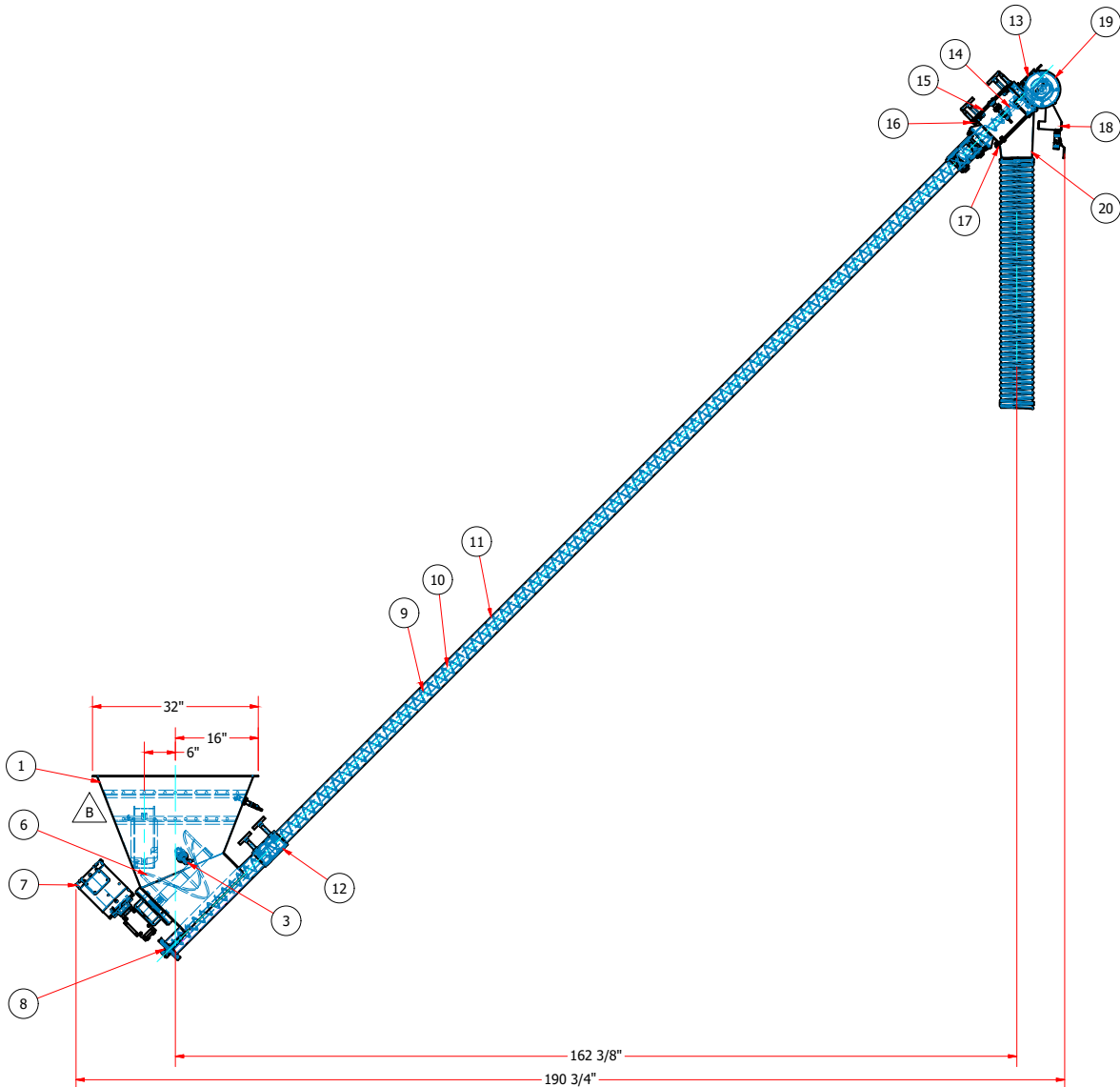
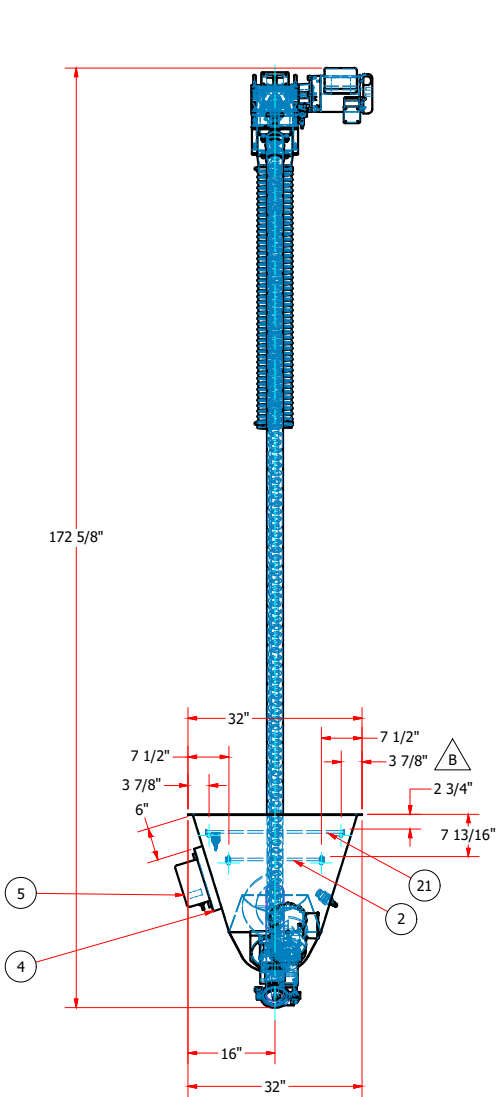
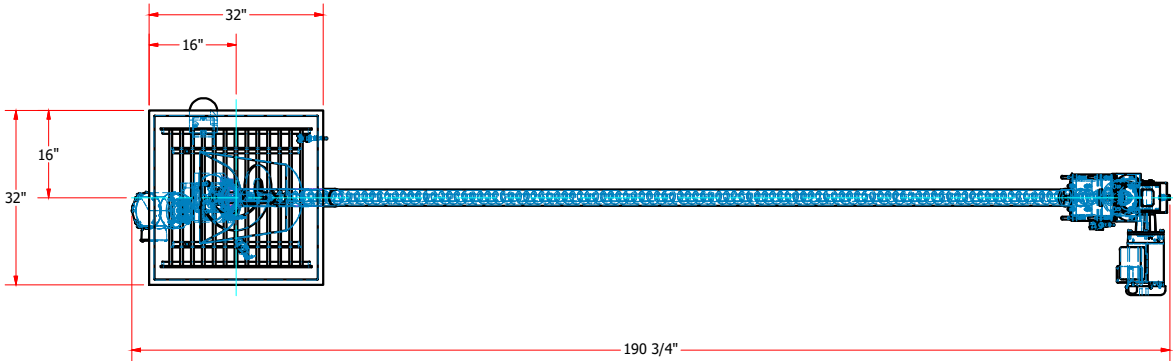
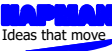


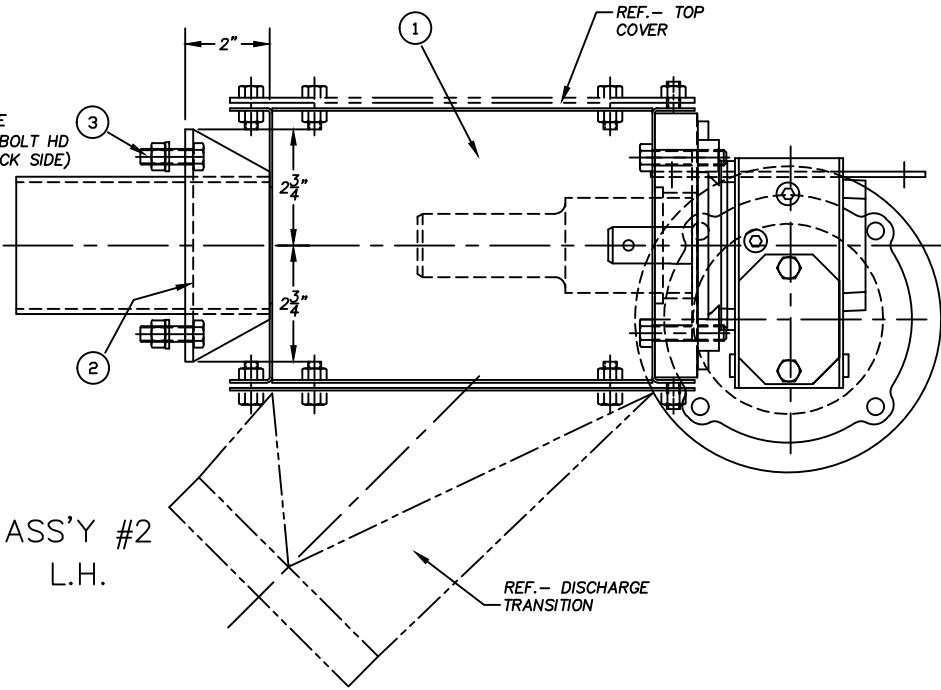
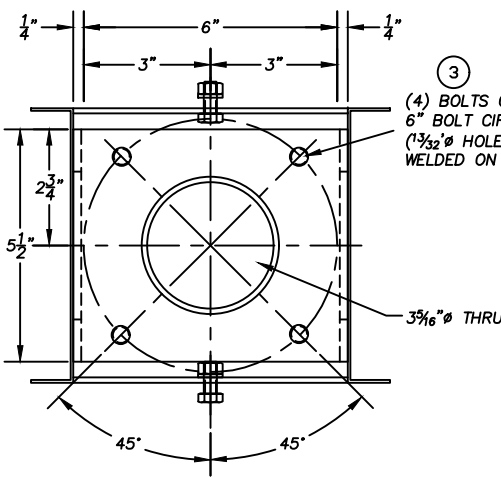
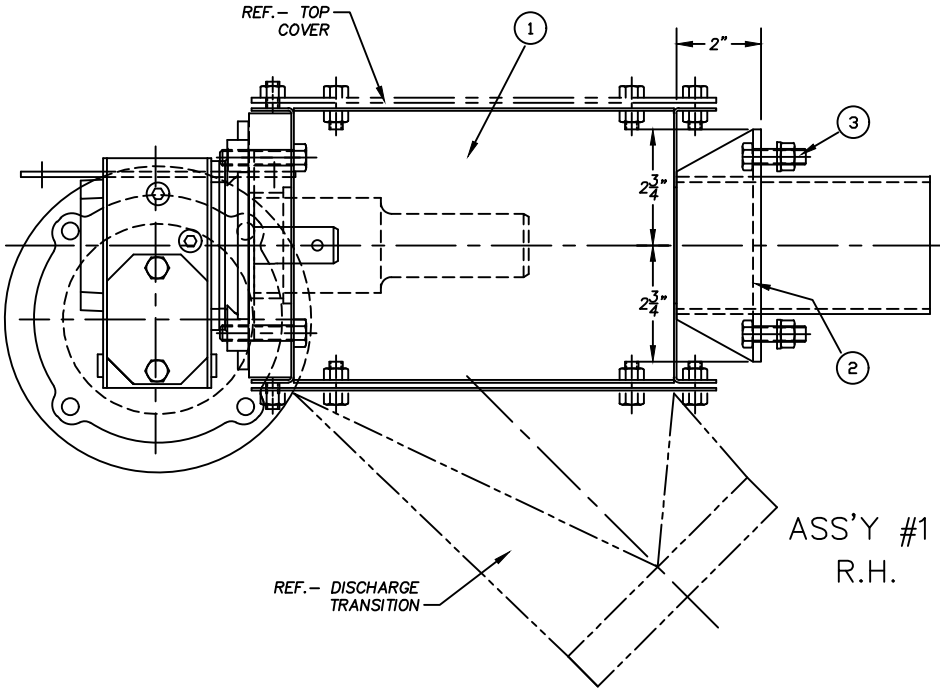
REVISION HISTORY			
REV	REVISION DESCRIPTION	DATE	BY
A	UPDATED OVERALL VIEWS TO SHOW LONGER FLEX HOSE	7/10/23	
B	UPDATED ITEM #2 TO DIFFERENT BAR GRATE, ADDED ITEM #21	8/2/23	LAT



REV	DATE	BY	DESCRIPTION	ITEM NO.
1			AGITATOR HOPPER ASSY, 30", 45 DEG, 304SS, FERRULE STYLE W 6" PROBE	1
2			BAR GRATE ASSEMBLY, BOLT IN, (LOW IN HOPPER), 304 SS	2
3			LOW LEVEL PROXIMITY SWITCH, 30-45 DEG UNIVERSAL HOPPERS	3
4			CHANNEL, VIBRATOR MTG. (SYNTRON V-20), SS	4
5			V-20	5
6			AGITATOR-RIBBON ASSY, 12", 200-800 SERIES HELIX, 3/4 HP, 224, 80:1, SS PLATE, SS	6
7			MOTOR, 3/4 HP, 230-460/3/60, 1800 RPM, 56C, WASH DOWN, PAINT FREE, TEFC, INVERTER DUTY	7
8			FERRULE CAP PROXIMITY SWITCH	8
9			AUGER BEVEL, 300 FLAT, 3/16" x 1 1/2" ID x 2 3/8" OD x 2" PITCH x 231" LG., SS	9
10			CORE, 1 3/8" OD x 1 1/8" ID x 1/8" WALL x 222" LG., POLY	10
11			POLY CASING, 2.96 OD x 2.7 ID x .130 WALL x 202" LG., POLY	11
12			COUPLING, STEPPED, 3 1/4" DIA. x 3" DIA., SS QUICK RELEASE (LORENZ) 325X3HDW	12
13			SERIES 300 HELIX, RIGHT ANGLE ASSY., 140 FRAME, RH, 5:1 RATIO, 360 RPM, 304 SS	13
14			SERIES 200 & 300 HELIX, U-BOLT, 304 SS (SHIP LOOSE W/DRIVE ASSY.)	14
15			250-300 SERIES, QUICK RELEASE DRIVE BOX COVER, 304 SS	15
16			GASKET PACKAGE, SERIES 200,300,400,500	16
17			HARDWARE PACKAGE, SS (SHIP LOOSE W/DRIVE ASSY.)	17
18			DRIVE DRIP PAN, 45 DEG INLINE DRIVE, #16 GA, SS	18
19			MOTOR, 2 HP, 230-460/3/60, 1800 RPM, 145TC, WASH DOWN, PAINT FREE, TEFC, INVERTER DUTY	19
20			DISCHARGE TRANSITION, 45 DEG x 6" OUTLET, SS	20
21			30" SQ. BAR GRATE W/ INTERLOCK, 304 SS	21

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE ENGLISH (FEET AND/OR INCHES) TOLERANCES			
DECIMALS	INCHES	FEET	ANGLES
.X ±.030"	X/X" = ± 1/16"	.X'-X" = ± 1/4" UP TO 20 FEET	° ± .1°
.XX ±.015"		.X'-X" = 1/2" OVER 20 FEET	
.XXX ±.005"			
DO NOT SCALE DRAWING		REMOVE ALL SHARP EDGES	
		5944 E. N AVE. KALAMAZOO, MICHIGAN 49048 PHONE (269) 343-1675 FAX (269) 349-2477	
DATE: 5/30/2023	DB NO.:	X20335	
SCALE:		DRAWN: lthompson	
TITLE: HELIX CONVEYOR ASSEMBLY		WEIGHT: N/A	
CUSTOMER:		DRAWING NO.:	
SHEET 1 OF 1		X20335AA-D00	
			B

REV	DESCRIPTION	DATE	BY
A	ITEM #3 WAS 3/8"-16 X 1" LG STUDS	12/22/10	DRB



DWG #	ITEM #	PART ID	DESCRIPTION	QTY.
9473-C01 ASS'Y 2, 360 RPM	1	X08728SD-C01	RIGHT ANGLE DR- 300, 140TC, 5:1 RATIO, SS (ASS'Y 2)	1
	2	.	CHANNEL, FORMED- 7GA, SS (SEE DETAIL)	1
	3	.	HX HD BOLT- 3/8-16 X 1.25 LG, SS, W/FWR & LOCK NUT	4

9473-C02 ASS'Y 2, 240 RPM	1	X08728SD-C02	RIGHT ANGLE DR- 300, 140TC, 7.5:1 RATIO, SS (ASS'Y 2)	1
	2	.	CHANNEL, FORMED- 7GA, SS (SEE DETAIL)	1
	3	.	HX HD BOLT- 3/8-16 X 1.25 LG, SS, W/FWR & LOCK NUT	4

9473-C03 ASS'Y 2, 180 RPM	1	X08728SD-C03	RIGHT ANGLE DR- 300, 140TC, 10:1 RATIO, SS (ASS'Y 2)	1
	2	.	CHANNEL, FORMED- 7GA, SS (SEE DETAIL)	1
	3	.	HX HD BOLT- 3/8-16 X 1.25 LG, SS, W/FWR & LOCK NUT	4

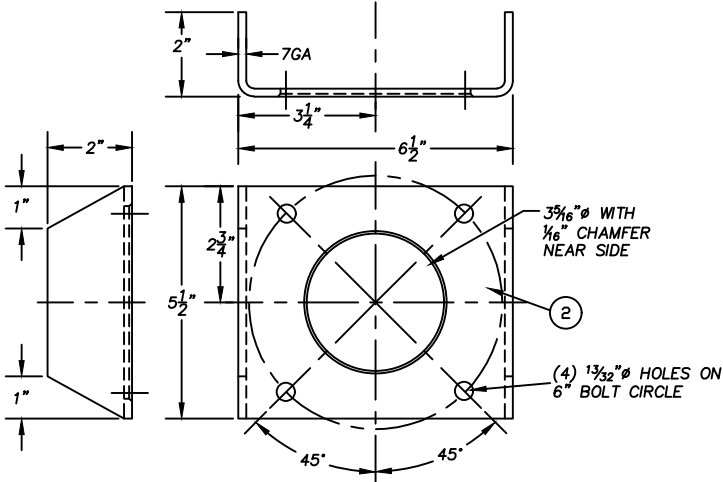
9473-C04 ASS'Y 1, 360 RPM	1	X08728SD-C04	RIGHT ANGLE DR- 300, 140TC, 5:1 RATIO, SS (ASS'Y 1)	1
	2	.	CHANNEL, FORMED- 7GA, SS (SEE DETAIL)	1
	3	.	HX HD BOLT- 3/8-16 X 1.25 LG, SS, W/FWR & LOCK NUT	4

9473-C05 ASS'Y 1, 240 RPM	1	X08728SD-C05	RIGHT ANGLE DR- 300, 140TC, 7.5:1 RATIO, SS (ASS'Y 1)	1
	2	.	CHANNEL, FORMED- 7GA, SS (SEE DETAIL)	1
	3	.	HX HD BOLT- 3/8-16 X 1.25 LG, SS, W/FWR & LOCK NUT	4

9473-C06 ASS'Y 1, 180 RPM	1	X08728SD-C06	RIGHT ANGLE DR- 300, 140TC, 10:1 RATIO, SS (ASS'Y 1)	1
	2	.	CHANNEL, FORMED- 7GA, SS (SEE DETAIL)	1
	3	.	HX HD BOLT- 3/8-16 X 1.25 LG, SS, W/FWR & LOCK NUT	4

9473-C07 ASS'Y 2, 120 RPM	1	X08728SD-C07	RIGHT ANGLE DR- 300, 140TC, 15:1 RATIO, SS (ASS'Y 2)	1
	2	.	CHANNEL, FORMED- 7GA, SS (SEE DETAIL)	1
	3	.	HX HD BOLT- 3/8-16 X 1.25 LG, SS, W/FWR & LOCK NUT	4

9473-C08 ASS'Y 1, 120 RPM	1	X08728SD-C08	RIGHT ANGLE DR- 300, 140TC, 15:1 RATIO, SS (ASS'Y 1)	1
	2	.	CHANNEL, FORMED- 7GA, SS (SEE DETAIL)	1
	3	.	HX HD BOLT- 3/8-16 X 1.25 LG, SS, W/FWR & LOCK NUT	4

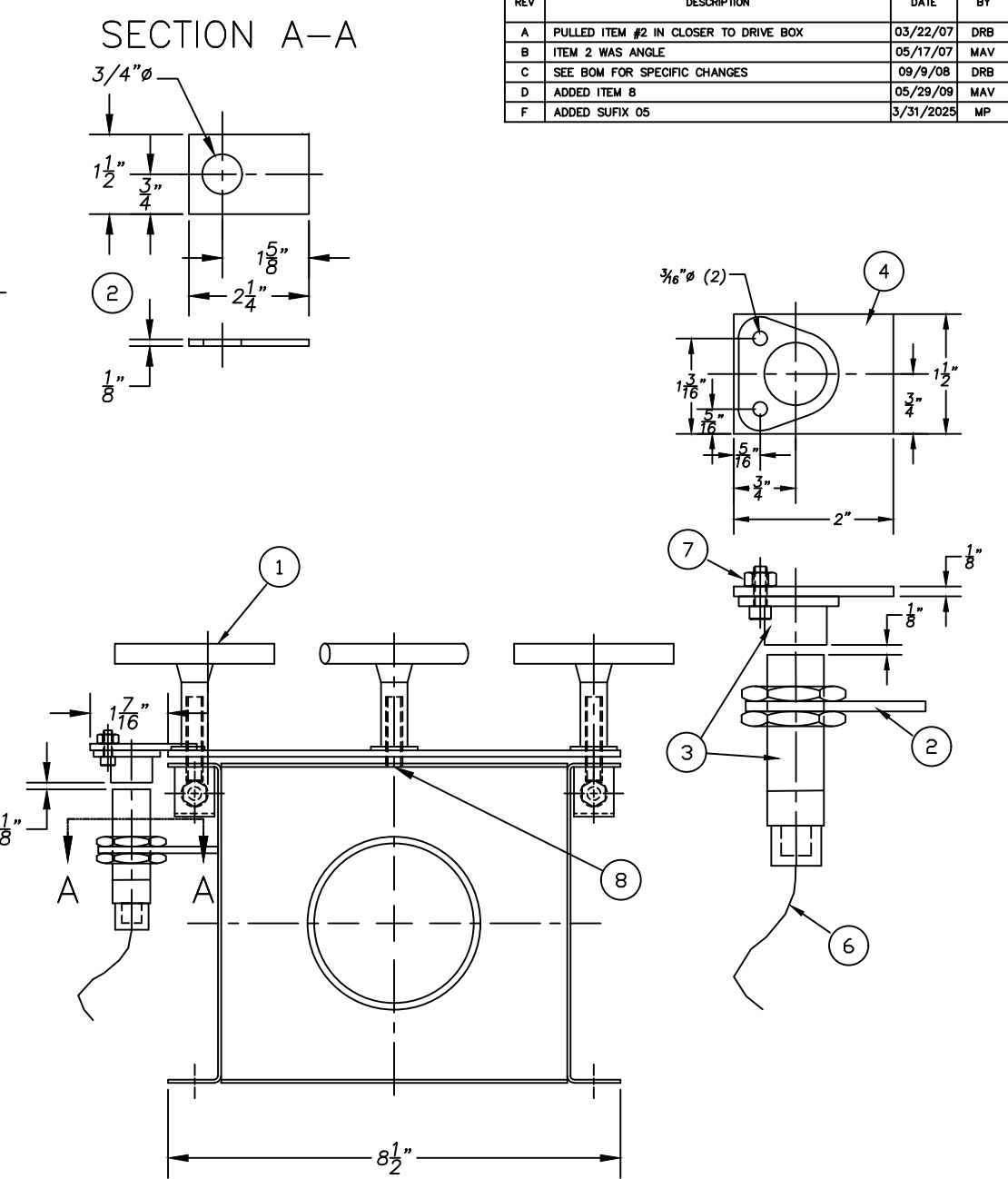


HAPMAN
Ideas that move

P.O. BOX 2321
6002 E. KILGORE ROAD, KALAMAZOO, MICHIGAN 49048
PHONE (269) 343-1675 FAX (269) 349-2477

DATE	4/2/09	JOB NO.		DRAWN	DRB
SCALE	1/2" = 1"			BOM =	1
				DWG =	1
TITLE					
HI/LO RIGHT ANGLE DRIVE ASSEMBLY					
300 SERIES, 140TC MOTORS, SS (CHARTED)					
CUST.					
SHEET	2	OF	2	DWG. NO.	9473-C__
				REV.	A

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE ENGLISH (FEET AND/OR INCHES) TOLERANCES			
DECIMALS	INCHES	FEET	ANGLES
.X ±.030			X'-X"=±1/4" UP TO 20 FEET
.XX ±.015			X'-X"=±1/2" OVER 20 FEET
.XXX ±.005			X'-X"=±1/16"
DO NOT SCALE DWG. BREAK ALL SHARP EDGES			
MATERIAL			
FINISH			



REV	DESCRIPTION	DATE	BY
A	PULLED ITEM #2 IN CLOSER TO DRIVE BOX	03/22/07	DRB
B	ITEM 2 WAS ANGLE	05/17/07	MAY
C	SEE BOM FOR SPECIFIC CHANGES	09/09/08	DRB
D	ADDED ITEM 8	05/29/09	MAY
F	ADDED SUFIX 05	3/31/2025	MP

05	304 SS	(ALLEN BRADLEY) SIPHA CONTROL W/S'STL SENSOR, WITH HAPMAN SUPPLIED CONTROLS
04	316 SS	(ALLEN BRADLEY) SIPHA CONTROL, SHIPPED LOOSE, CUSTOMER SUPPLIED CONTROLS
03	316 SS	(ALLEN BRADLEY) SIPHA CONTROL, WITH HAPMAN SUPPLIED CONTROLS
02	304 SS	(ALLEN BRADLEY) SIPHA CONTROL, SHIPPED LOOSE, CUSTOMER SUPPLIED CONTROLS
01	304 SS	(ALLEN BRADLEY) SIPHA CONTROL, WITH HAPMAN SUPPLIED CONTROLS

HAPMAN
Ideas that move

P.O. BOX 2321
6002 E. KILGORE ROAD, KALAMAZOO, MICHIGAN 49048
PHONE (269) 343-1675 FAX (269) 349-2477

DATE	8/26/05	JOB NO.	DRAWN		DRB
SCALE	1/2" = 1"		BOM = 1	DWG = 1	

QUICK RELEASE DRIVE BOX
COVER ASSEMBLY- (300 & 250 SERIES)

CUST.		
SHEET 2 OF 2	DWG. NO. 9179-C__	REV. F

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE ENGLISH (FEET AND/OR INCHES) TOLERANCES			
DECIMALS	INCHES	FEET	ANGLES
.XX ± .030	$X'' - X'' = \pm 1/16''$	$X' - X' = \pm 1/4''$ UP TO 20 FEET $X' - X' = \pm 1/2''$ OVER 20 FEET	± 1°
.X ± .015 .XXX ± .005			
DO NOT SCALE DWG. BREAK ALL SHARP EDGES			
MATERIAL			
FINISH			



Description

When it comes to machine safety, Rockwell Automation knows that protection of personnel and equipment is your main concern. At the same time, flexibility and productivity are points that must also be considered as you design your safety system. Optimize all of these with the new Allen-Bradley SensaGuard family of noncontact switches.

Featuring the latest generation of RFID technology for coding and inductive technology for sensing, SensaGuard's large sensing range and tolerance to misalignment is a cost-effective solution that is ideally suited for a wide range of industrial safety applications.

The SensaGuard product line is a Category 4 /SIL 3 rated switch per EN954-1, TÜV functional safety approved to IEC 61508.

Features

- Switches can be connect to a standard safety relay, for example, the MSR126, MSR127, MSR200/300 Family, SmartGuard and Safety I/O Blocks
- Multiple actuator sizes for large sensing distance
- IP 69K environmental rating
- Short-circuit and over-voltage protection
- Led located on the switch for door status and troubleshooting

Benefits

- No dedicated controller required
- Cat 4/SIL 3 rating maintained even with multiple units connected in series
- Switches can be connected in series with other devices (light curtain, E-stops, key interlock switches)
- Extended diagnostics for easy troubleshooting
- Large sensing distances
- Tolerance to misalignment
- Multiple sensing directions
- Stainless steel version suitable for use in harsh environments
- Use standard proximity brackets

Specifications

Category	Cat. 4/SIL3		
Certifications	TÜV, CE, cULus (UL 508)		
Standards	IEC 60947-5-3, IEC 61508, EN 954		
PFHd	1.119...10 ⁻⁹		
Operating Characteristics			
Sensing Distance (Target)	18 mm Plastic	30 mm Plastic	18 mm SS
Operating Distance, Make—mm (in)	Assured: 15 (0.59)	Assured: 25 (0.98)	Assured: 10 (0.39)
Case Material	Valox® DR 48		304 Stainless Steel
Actuator Material	Valox® DR 48		304 Stainless Steel
Typical Misalignment	See misalignment curve.		
Repeat Accuracy	10% of Sensing Range		
Output Current, Max.	200 mA (all outputs)		
Switching Current @ Voltage, Max.	24V DC +10%/-15%		
Operating Voltage/Supply Current	24V DC, +10%/-15% Class 2 SELV power supply		
Frequency of Operating Cycle	1 Hz		
Response Time (Off)	54 ms		
Outputs (Guard Door Closed, Actuator in Place)			
Safety Outputs	2 x PNP, 0.2 A, max.; Status: ON (+24V DC)		
Auxiliary Outputs	1 x PNP, 0.2 A max.; Status: OFF (0V DC)		
Environmental			
Operating Temperature—C (F)	-10...+55° (+14...+131°)		
Relative Humidity	5...95%		
Enclosure Type Rating	NEMA 3, 4X, 12, 13, IP 69K		
Shock	IEC68-2-27 30 g, 11 ms		
Vibration	IEC 68-2-6 10...55 Hz		
Radio Frequency	IEC 61000-4-3, IEC 61000-4-6		
Protection			
Protection Type	Short Circuit, Current Limitation, Overload, False Pulse, Transient Noise, Reverse Polarity, Overvoltage (inc. load dump), Thermal Shutdown with restart, Loss of GND and Vdd		
Loss of GND and Vdd protection	Incorporated		
Electrical Life	10 x 10 ⁶		

Safety Switches
Noncontact Switches
 SensaGuard™ 18 mm Barrel

Product Selection

Description	Operating Voltage/ Input Current	Safety Outputs	Auxiliary Outputs	Actuator Type	Assured Sensing Distance	Connection	Cat. No.
18 mm Plastic Barrel	24V DC, +10%/-15%/ 50 mA max.	2 Solid State outputs	1 Solid State output	18 mm	15 mm (0.59 in)	6 in Pigtail, 8-Pin Micro (M12)	440N-Z21S16H
						3 m Cable	440N-Z21S16A
						10 m Cable	440N-Z21S16B
				30 mm	25 mm (0.98 in)	6 in Pigtail, 8-Pin Micro (M12)	440N-Z21S26H
						3 m Cable	440N-Z21S26A
						10 m Cable	440N-Z21S26B
18 mm Stainless Steel Barrel	18 mm	10 mm (0.39 in)	6 in Pigtail, 8-Pin Micro (M12)	440N-Z21S17H			
			3 m Cable	440N-Z21S17A			
			10 m Cable	440N-Z21S17B			

Recommended Logic Interfaces

Description	Safety Outputs	Auxiliary Outputs	Terminals	Reset Type	Power Supply	Cat. Pg. No.	Cat. No.
Single-Function Safety Relays							
MSR127RP	3 N.O.	1 N.C.	Removable Screw	Monitored Manual	24V AC/DC	—	440R-N23135
MSR127TP	3 N.O.	1 N.C.	Removable Screw	Automatic/Manual	24V AC/DC	—	440R-N23132
Modular Safety Relays							
MSR211P Base 2 N.C. only	2 N.O.	1 N.C. and 2 PNP Solid State	Removable	Automatic/Manual Monitored Manual	24V DC	—	440R-H23176
MSR220P Input Module	—	—	Removable	—	24V DC	—	440R-H23178
MSR310P Base	—	3 PNP Solid State	Removable	Automatic/Manual Monitored Manual	24V DC	—	440R-W23219
MSR320P Input Module	—	2 PNP Solid State	Removable	—	24V DC	—	440R-W23218

Note: For additional Safety Relays connectivity, see the Logic section of this catalog.
 For additional Safety I/O connectivity, see the Safety I/O section of this catalog.
 For Application and wiring diagrams, see the Applications section of this catalog.

Connection Systems

Description	Cat. No.
Cordset	889D-F8AB-*

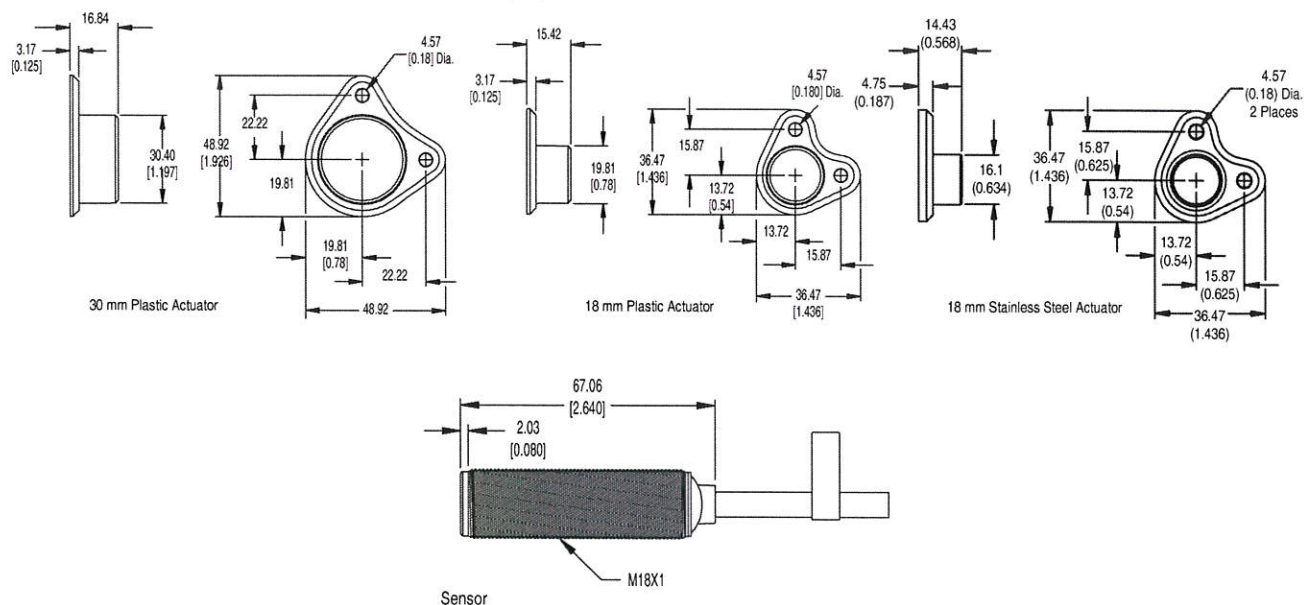
* Replace symbol with 2 (2 m), 5 (5 m), or 10 (10 m) for standard cable lengths.

Accessories

	Description	Dimensions	Cat. No.
	18 mm Plastic Actuator	—	440N-Z18PT
	30 mm Plastic Actuator	—	440N-Z30PT
	18 mm Stainless Steel Actuator	—	440N-Z18SST
	Mounting Bracket for Tubular Sensors—Right Angle Style	—	871A-BRS18
	Mounting Bracket for Tubular Sensors—Clamp Style	—	871A-BP18
	Snap Clamp Mounting Bracket	—	871A-SCBP18
	Swivel/Tilt bracket allows $\pm 10^\circ$ vertical and 360° rotation adjustment.	—	60-2649

Approximate Dimensions—mm (inches)

Dimensions are not intended to be used for installation purposes.

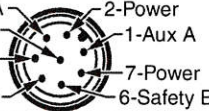


Safety Switches

Noncontact Switches

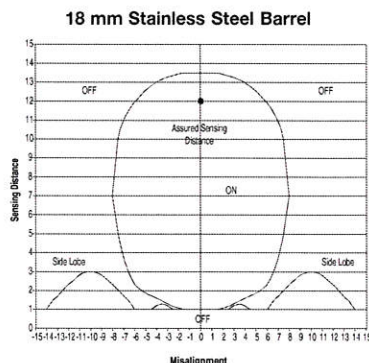
SensaGuard™ 18 mm Barrel

Typical Wiring Diagrams

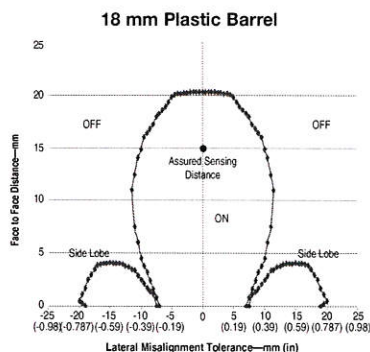
Description		Plastic		Stainless Steel	
8-Pin Micro (M12)					
8-Pin Cordset 889D-F8AB-*	Grey	OSSD 1	Safety A	OSSD 1	Safety A
	Red	OSSD 1+		OSSD 1+	
	Pink	OSSD 2	Safety B	OSSD 2	Safety B
	Yellow	OSSD 2+		OSSD 2+	
	White	Aux A		Aux A	
	Brown	24V DC	Unit Power	24V DC	Unit Power
	Blue	Gnd		Gnd	
	Green	NA		Shield	

* Replace symbol with 2 (2 m), 5 (5 m) or 10 (10 m) for standard cable lengths.

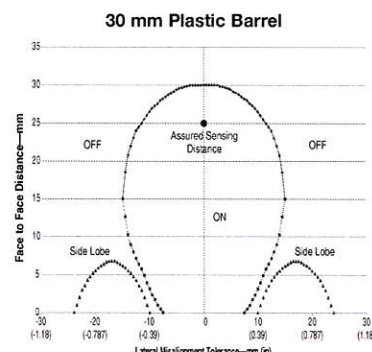
Misalignment Curve



Note: There must be a minimum spacing of 4 mm (0.157 in) if actuator and sensor face approaches laterally. This will prevent false triggering due to the side lobe areas.

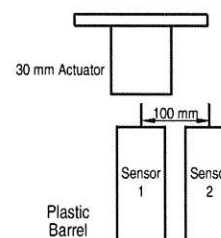
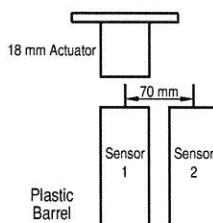
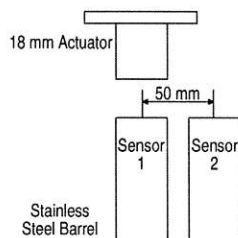


Note: There must be a minimum spacing of 4 mm (0.157 in) if actuator and sensor face approaches laterally. This will prevent false triggering due to the side lobe areas.



Note: There must be a minimum spacing of 7 mm (0.275 in) if actuator and sensor face approaches laterally. This will prevent false triggering due to the side lobe areas.

Minimum Distance Between Sensors



Diagnostic

Unit Indicators (per IEC 60073)

Device Output LED	State	Status	Troubleshooting
	Off	Not Powered	NA
	Red	Not Safe, OSSD Not Active	NA
	Green	Safe, OSSD Active	NA
	Green Flash	Power Up Test or OSSD Inputs Not Valid	Check 24V DC on OSSD Inputs (yellow and red wire)
	Red Flash	1 Hz Flash Recoverable Fault 4 Hz Flash Nonrecoverable Fault	Recoverable Fault: Check OSSD Outputs Are Not Shorted to GND, 24V DC or Each Other. Cycle Power.